$$\left\{\text{Only 1-loops} = \frac{1}{3}\times\begin{pmatrix} 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 \end{pmatrix},\right.$$

$$\text{Only 1- and 2-loops} = \frac{1}{3}\times\begin{pmatrix} 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 \end{pmatrix},$$

$$\text{1- and 2-loops and a 1-cycle} = \frac{1}{3}\times\begin{pmatrix} 6 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -3 & 0 & 0 & 0 & 3\omega & 3\omega^2 & 0 & 0 \\ 0 & 0 & -3 & 3\omega^2 & 0 & 0 & 0 & 3\omega & 0 \\ 0 & 0 & 3\omega & -3 & 0 & 0 & 0 & 3\omega^2 & 0 \\ 3 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 3 \\ 0 & 3\omega^2 & 0 & 0 & 0 & -3 & 3\omega & 0 & 0 \\ 0 & 3\omega & 0 & 0 & 0 & 3\omega^2 & -3 & 0 & 0 \\ 0 & 0 & 3\omega^2 & 3\omega & 0 & 0 & 0 & -3 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 6 \end{pmatrix},$$

$$\text{1- and 2-loops and a 1-cycle and a 2-cycle in the other direction} =$$

$$\frac{1}{3}\times\begin{pmatrix} 6 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -3 & 0 & 0 & 0 & 3\omega & 3\omega^2 & 0 & 0 \\ 0 & 0 & -3 & 3\omega^2 & 0 & 0 & 0 & 3\omega & 0 \\ 0 & 0 & 3\omega & -3 & 0 & 0 & 0 & 3\omega^2 & 0 \\ 3 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 3 \\ 0 & 3\omega^2 & 0 & 0 & 0 & -3 & 3\omega & 0 & 0 \\ 0 & 3\omega & 0 & 0 & 0 & 3\omega^2 & -3 & 0 & 0 \\ 0 & 0 & 3\omega^2 & 3\omega & 0 & 0 & 0 & -3 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 6 \end{pmatrix},$$

$$\text{Everything except 0-loops} = \frac{1}{3}\times\begin{pmatrix} 6 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 9 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 9 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 9 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 6 \end{pmatrix},$$

$$\text{One 1-cycle} = \frac{1}{3}\times\begin{pmatrix} 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 3\omega & 3\omega^2 & 0 & 0 \\ 0 & 0 & 0 & 3\omega^2 & 0 & 0 & 0 & 3\omega & 0 \\ 0 & 0 & 3\omega & 0 & 0 & 0 & 0 & 3\omega^2 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & 3\omega^2 & 0 & 0 & 0 & 0 & 3\omega & 0 & 0 \\ 0 & 3\omega & 0 & 0 & 0 & 3\omega^2 & 0 & 0 & 0 \\ 0 & 0 & 3\omega^2 & 3\omega & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\text{All 1's}^2 <> = \frac{1}{3}\times\left.\begin{pmatrix} 6 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 6 \\ 0 & -3 & 0 & 0 & 0 & -3 & -3 & 0 & 0 \\ 0 & 0 & -3 & -3 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & -3 & -3 & 0 & 0 & 0 & -3 & 0 \\ 6 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 6 \\ 0 & -3 & 0 & 0 & 0 & -3 & -3 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & -3 & -3 & 0 & 0 \\ 0 & 0 & -3 & -3 & 0 & 0 & 0 & -3 & 0 \\ 6 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 6 \end{pmatrix}\right\}$$